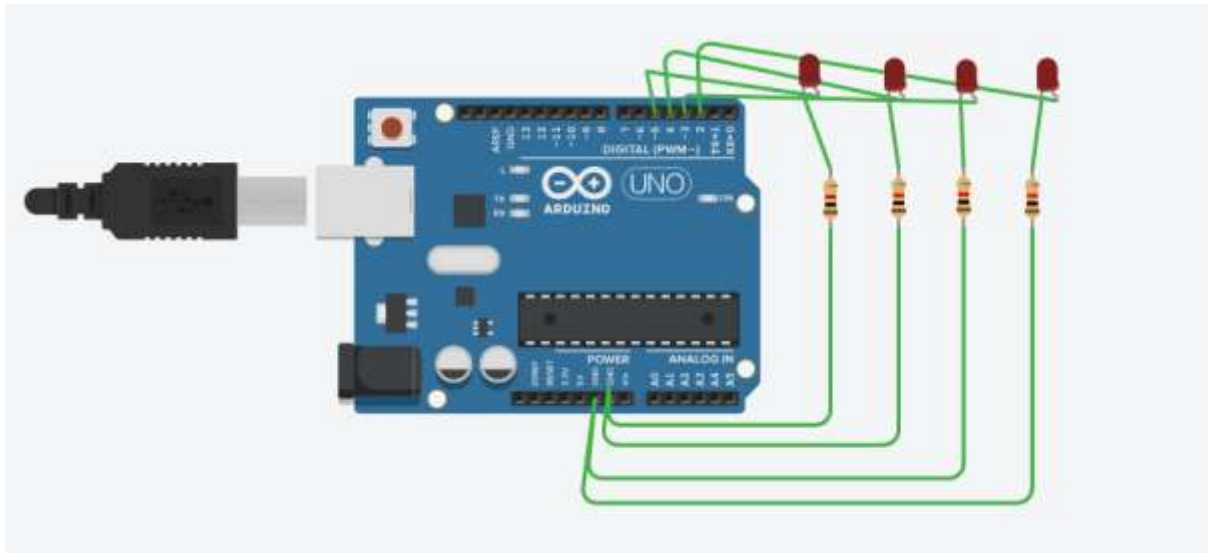




```
int LED1 = 5;
int LED2 = 4;
int LED3 = 3;
int LED4 = 2;
void setup() {
  Serial.begin(9600);
  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);
  pinMode(LED3, OUTPUT);
  pinMode(LED4, OUTPUT);
  Serial.println("Enter a decimal number (0 to 15):");
}
void loop() {
  if (Serial.available() > 0) {
    int decimalVal = Serial.parseInt();
    if (decimalVal >= 0 && decimalVal <= 15) {
      Serial.print("Decimal: ");
      Serial.print(decimalVal);
```

```
Serial.print(" -> Binary: ");
int bit3 = (decimalVal >> 3) & 1;
int bit2 = (decimalVal >> 2) & 1;
int bit1 = (decimalVal >> 1) & 1;
int bit0 = (decimalVal >> 0) & 1;
Serial.print(bit3);
Serial.print(bit2);
Serial.print(bit1);
Serial.println(bit0);
digitalWrite(LED1, bit3 ? HIGH : LOW);
digitalWrite(LED2, bit2 ? HIGH : LOW);
digitalWrite(LED3, bit1 ? HIGH : LOW);
digitalWrite(LED4, bit0 ? HIGH : LOW);
} else {
    Serial.println("Error: Please enter a valid number between 0 and 15.");
}
while (Serial.available() > 0) {
    Serial.read();
}
}
}
```